

HOMework 3, EXTRA CREDIT PORTION

(LINEAR ALGEBRA REVIEW)

NUMERICAL ANALYSIS
(MATH 461) FALL, 2023

DUE: 09/22/2023

Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

Problem 1. Elementary row operation on an augmented matrix never change the solution set of the associated linear system.

☐ True☐ False

Problem 2. The reduced row echelon form of a matrix is unique.

☐ True☐ False

Problem 3. A homogeneous system is always consistent.

☐ True☐ False

Problem 4. If A , B , and C are matrices such that $AB = AC$, then $B = C$.

☐ True☐ False

Problem 5. The columns of any 4×5 matrix are linearly dependent.

☐ True☐ False

Problem 6. Every matrix transformation is a linear transformation.

☐ True☐ False

Problem 7. A linear transformation preserves the operations of vector addition and scalar multiplication.

☐ True☐ False

Problem 8. If two vectors are linearly independent, then they both lie on the same line through the origin.

☐ True☐ False

Problem 9. If A and B are $n \times n$ matrices such that $AB = I$, then A and B are invertible.

☐ True☐ False

Problem 10. A column of the standard matrix for a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the image of a column of the $n \times n$ identity matrix.

☐ True

☐ False

Problem 11. A subspace of $\mathbb{R}^n$ is any set H contained in $\mathbb{R}^n$ such that:

(1) $0 \in H$, (2) H is closed under vector addition, (3) H is closed under scalar multiplication.

☐ True

☐ False

Problem 12. The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$.

☐ True

☐ False

Problem 13. The value of $\text{rank}(A)$ is the dimension of $\text{Nul}(A)$.

☐ True

☐ False

Problem 14. If $\mathcal{B} = \{v_1, \dots, v_p\}$ is a basis for a subspace H , and if $x = c_1 v_1 + \dots, c_p v_p$ for some $x \in H$, then the values of $c_1, \dots, c_p$ are the entries in the $\mathcal{B}$ -coordinate vector $[x]_{\mathcal{B}}$.

☐ True

☐ False

Problem 15. If $\mathcal{B} = \{v_1, \dots, v_p\}$ is a basis for a subspace H , then any vector x in H can be written in at least two different ways as a linear combination of the vectors in $\mathcal{B}$.

☐ True

☐ False

Problem 16. If A is a triangular $n \times n$ matrix, then $\det(A)$ is the product of the entries on the main diagonal of A .

☐ True

☐ False

Problem 17. If the columns of A are linearly dependent, then $\det(A) = 0$.

☐ True

☐ False

Problem 18. The set of all solutions to a homogeneous linear system is the kernel of a linear transformation.

☐ True

☐ False

Problem 19. If B is an echelon form of a matrix A , then the pivot columns of B form a basis for $\text{Col}(A)$.

☐ True

☐ False

Problem 20. If A is an invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

☐ True

☐ False

Problem 21. If λ is a scalar such that $Ax = \lambda x$ for some $x \in \mathbb{R}^n$, λ is an eigenvalue of A .

☐ True

☐ False

Problem 22. $\det(AB) = \det(A) - \det(B)$.

☐ True

☐ False

Problem 23. If $A = P \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} P^{-1}$, then $A^{2019} = P \begin{bmatrix} 3^{2019} & 0 \\ 0 & 8^{2019} \end{bmatrix} P^{-1}$.

☐ True

☐ False

Problem 24. If A is diagonalizable, then A is invertible.

☐ True

☐ False

Problem 25. If a vector x is orthogonal to every vector in a subspace W , then $x \in W^\perp$.

☐ True

☐ False

Problem 26. If A is any $n \times m$ matrix, then $(\text{Col}(A))^\perp = \text{Nul}(A)$.

☐ True

☐ False

Problem 27. If $\{v_1, \dots, v_p\}$ is a linearly independent set in $\mathbb{R}^n$, then it is also an orthogonal set.

☐ True

☐ False

Problem 28. If $\{v_1, \dots, v_p\}$ is an orthogonal basis for a subspace W of $\mathbb{R}^n$, then for any $y \in W$, we have $y = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{y \cdot v_p}{v_p \cdot v_p} v_p$.

☐ True

☐ False

Problem 29. The best approximation to y by elements of a subspace W is given by the vector $y - \text{proj}_W(y)$.

☐ True

☐ False

Problem 30. If $x \in \mathbb{R}^n$ is a nonzero vector and W is a nonzero subspace of $\mathbb{R}^n$, then $\text{proj}_W(x) \neq 0$.

☐ True

☐ False